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Original Report

# Effect of high-dose stereotactic body radiation therapy on liver function in the treatment of primary and metastatic liver malignancies using the Child-Pugh score classification system

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**Abstract**

**Purpose:** The purpose of this study was to evaluate liver function after high-dose liver stereotactic body radiation therapy (SBRT) in the treatment of metastatic and primary malignancies of the liver using the Child-Pugh score classification system.

**Methods and materials:** This was a retrospective analysis of 46 patients treated with SBRT for metastatic and primary malignancies of the liver. Patient, disease, prior treatment, and SBRT dosimetric factors were analyzed to correlate with decline in Child-Pugh class after liver SBRT.

**Results:** Median follow-up was 11.0 months for patients alive at last follow-up. Twenty-three patients (50%) had primary liver malignancies. Median delivered dose was 55 Gy in 5 fractions (range, 36-60 Gy in 3-6 fractions) to 1 lesion (range, 1-4 lesions) measuring 4.0 cm (range, 1.3-12.4 cm). Forty-one patients (89%) received $\geq$50 Gy in 3 to 6 fractions. Child-Pugh score classification was A in 42 patients (91%). Seven patients (15%) received adjuvant chemotherapy or targeted therapy. Twenty-nine patients (63%) experienced an intrahepatic recurrence after treatment. Ten patients (22%) experienced a decline in Child-Pugh class at a median of 1.6 months (range, 0.2-6 months). Eighty percent experienced a one-category decline. Only the V20, V25, V30, and V50 were correlated with decline in Child-Pugh class on univariate analysis, with V25 being most significant ($P$ = .027). A V25 >32% was associated with a 42% incidence of Child-Pugh class decline compared with 9% for V25 $\leq$32 ($P$ = .029). For primary liver malignancies, a V25 >36% was associated with a 4-fold increase in the incidence of Child-Pugh class decline (60% vs 15%, $P$ = .021).

Supplementary material for this article (http://dx.doi.org/10.1016/j.prro.2014.09.007) can be found at www.practicalradonc.org.

Conflicts of interest: Dr Olsen provided consulting regarding the use of palliative radiation therapy for bone metastases to DFINE, Inc and received research support from RadioMed, Inc, honoraria from ViewRay, Inc, and travel reimbursement from RSNA Resident & Fellow Committee. Dr Parikh currently has grants/grants pending through Varian Medical Systems, ViewRay, Inc, and Philips Healthcare, has received payment for lectures including service on speakers' bureaus for Varian Medical Systems, and is a stockholder in Holaira, Inc. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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